

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405589
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Kurita; Kazuki et al.

Programmable logic controller, sequence control system, and non-transitory computer readable medium

Abstract

In a programmable logic controller, a flag data update receiver receives an update of flag data indicating whether activation of a second execution element is permitted or prohibited. An interrupt receiver receives an interrupt instruction indicating a request for activating the second execution element independently of whether a transition condition is satisfied. An interrupt preventer maintains, when the interrupt instruction is received and latest flag data indicates that the activation of the second execution element is prohibited, an inactive state of the second execution element and outputs an interrupt invalidation error indicating that the interrupt instruction is invalidated. An executor activates the second execution element when the interrupt instruction is received and the latest flag data indicates that the activation of the second execution element is permitted.

Inventors:	Kurita; Kazuki (Nagoya, JP), Nagatomo; Yoshihiro (Nagoya, JP)
Applicant:	Mitsubishi Electric Corporation (Tokyo, JP)
Family ID:	85980428
Assignee:	MITSUBISHI ELECTRIC CORPORATION (Tokyo, JP)
Appl. No.:	18/880285
Filed (or PCT Filed):	July 19, 2022
PCT No.:	PCT/JP2022/028025
PCT Pub. No.:	WO2024/018515
PCT Pub. Date:	January 25, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250164958 A1	May. 22, 2025

Publication Classification

Int. Cl.: G05B19/05 (20060101)

U.S. Cl.:

CPC G05B19/05 (20130101); G05B2219/13076 (20130101)

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4930068	12/1989	Katayose	710/261	G06F 13/24
2004/0088462	12/2003	Miyake	710/261	G06F 9/4812
2008/0046621	12/2007	Okino	710/260	G06F 9/4812
2009/0158260	12/2008	Moon et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1577291	12/2004	CN	N/A
H10-171527	12/1997	JP	N/A
2003-280923	12/2002	JP	N/A
2006-099331	12/2005	JP	N/A
20130137478	12/2012	KR	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed on Oct. 11, 2022, received for PCT Application PCT/JP2022/028025, filed on Jul. 19, 2022, 11 pages including English Translation. cited by applicant

Office Action dated Jun. 10, 2025, issued for the corresponding CN patent application No. 1 English translation, 17pp. cited by applicant

Primary Examiner: Ogg; David Earl

Attorney, Agent or Firm: XSENSUS LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a national stage application, pursuant to 35 U.S.C. § 371, of International Patent Application No. PCT/JP2022/028025, filed Jul. 19, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a programmable logic controller, a sequence control system, and an active state control program.

BACKGROUND ART

(3) A known programmable logic controller (PLC) controls devices that are involved in manufacturing on a production line. The PLC executes a sequence program to control the devices.

(4) The sequence program is written in a known programming language such as the sequential function chart. The sequence program written in the sequential function chart includes elements called steps and elements called transitions.

(5) The steps describe actions for a control target. The transitions describe transition conditions for transition from a preceding step (hereafter referred to as a first step) to the step immediately after the first step (hereafter referred to as a second step).

(6) A step includes two states, or an active state and an inactive state. The step in the active state refers to the step being performed. The step in the inactive state refers to the step not being performed. When a transition condition defined in a transition is satisfied, the active state switches from the first step to the second step.

(7) Patent Literature 1 describes a PLC that can maintain the inactive state of the second step to allow debugging the second step. The PLC includes flag data indicating whether activation of the second step is permitted or prohibited, and maintains the inactive state of the second step when the transition condition is satisfied but the flag data indicates that the activation of the second step is prohibited.

CITATION LIST

Patent Literature

(8) Patent Literature 1: Unexamined Japanese Patent Application Publication No. H10-171527

SUMMARY OF INVENTION

Technical Problem

(9) An execution element hereafter refers to an element defining, in the sequence program, one or more associated actions for the control target. The step is an example of the execution element. A transition condition element hereafter refers to an element defining a transition condition for transition from an execution element to the next execution element. The transition is an example of the transition condition element.

(10) The execution element can be activated by a factor other than the transition condition being satisfied. More specifically, the execution element can be activated by an interrupt instruction indicating a request for forcibly activating the execution element independently of whether the transition condition is satisfied.

(11) Examples of the interrupt instruction include an external interrupt request that is the request to the PLC from outside the PLC and an internal interrupt request that is the request generated internally in the PLC.

(12) The technique in Patent Literature 1 can prevent activation of the execution element due to the transition condition being satisfied, but cannot prevent activation of the execution element due to the interrupt instruction. The execution element that defines an action for a device may be activated by the interrupt instruction during maintenance of the device, and may cause a failure in the device. Under such circumstances, a technique is awaited for maintaining the inactive state of the execution element when the interrupt instruction is generated.

(13) An objective of the present disclosure is to provide a programmable logic controller, a sequence control system, and an active state control program that can maintain the inactive state of an execution element when an interrupt instruction is generated.

Solution to Problem

(14) A programmable logic controller according to an aspect of the present disclosure executes a sequence program including a first execution element defining a preceding action, a second execution element defining a subsequent action to be performed subsequently to the preceding

action, and a transition condition element defining a transition condition for transition of an active state from the first execution element to the second execution element. The programmable logic controller includes a flag data update receiver to receive an update of flag data indicating whether activation of the second execution element is permitted or prohibited, an interrupt receiver to receive an interrupt instruction indicating a request for activating the second execution element independently of whether the transition condition is satisfied, an interrupt preventer to refer to latest flag data when the interrupt receiver receives the interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, an inactive state of the second execution element and output an interrupt invalidation error indicating that the interrupt instruction is invalidated, and an executor to activate the second execution element when the interrupt receiver receives the interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.

Advantageous Effects of Invention

(15) The above structure can maintain the inactive state of the second execution element when the interrupt instruction is generated but the flag data indicates that the activation of the second execution element is prohibited.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a conceptual diagram of a sequence control system according to an embodiment;
- (2) FIG. 2 is a conceptual block diagram of a programmable logic controller (PLC) according to the embodiment;
- (3) FIG. 3 is a conceptual diagram of a sequence program according to the embodiment;
- (4) FIG. 4 is a functional conceptual diagram of the PLC according to the embodiment;
- (5) FIG. 5 is a flowchart of a transition prevention process in the embodiment;
- (6) FIG. 6 is a flowchart of an external interrupt prevention process in the embodiment; and
- (7) FIG. 7 is a flowchart of an internal interrupt prevention process in the embodiment.

DESCRIPTION OF EMBODIMENTS

(8) A sequence control system according to an embodiment is described below with reference to the drawings. Like reference signs denote like or corresponding components in the drawings.

(9) As illustrated in FIG. 1, a sequence control system **400** according to the present embodiment includes multiple devices **300** involved in manufacturing on a production line, a programmable logic controller (PLC) **100** that controls the devices **300**, and a programmable display **200** connected to the PLC **100** through a communication line CL.

(10) FIG. 1 illustrates devices **300A** to **300F** among the multiple devices **300**. The devices **300A** to **300F** are arranged along a conveyor CV that transports a workpiece. The workpiece herein includes a semifinished product under manufacturing and a component included in a product. Each of the devices **300A** to **300F** performs processing of the workpiece, such as attaching other components to, machining, marking, inspecting, or packaging the workpiece.

(11) The conveyor CV includes a main line CVa and a selection line CVb arranged parallel to the main line CVa. The selection line CVb has, in a direction in which the workpiece is transported, an upstream end that branches from the main line CVa. The selection line CVb has, in the direction in which the workpiece is transported, a downstream end that merges with the main line CVa.

(12) The device **300A**, the device **300B**, the device **300C**, and the device **300F** are arranged in this order along the main line CVa. The device **300D** and the device **300E** are arranged in this order along the selection line CVb.

(13) A manufacturing process performed by the device **300B** and the device **300C** is hereafter referred to as a process A, and a manufacturing process performed by the device **300D** and the

device **300E** as a process B. The process A and the process B are performed alternatively. In other words, when the device **300B** and the device **300C** operate, the device **300D** and the device **300E** stop. When the device **300D** and the device **300E** operate, the device **300B** and the device **300C** stop.

(14) The programmable display **200** functions as a human machine interface for the PLC **100**. In other words, the programmable display **200** is an example of a human machine device in an aspect of the present disclosure.

(15) The programmable display **200** includes a display screen **210** that serves as a graphical user interface. The display screen **210** is a touchscreen. A user operates the display screen **210** in the programmable display **200** to control the devices **300** and the conveyor CV through the PLC **100**.

(16) More specifically, the user can operate the display screen **210** in the programmable display **200** in the present embodiment to instruct the PLC **100** to switch from a state in which the process A is performed to a state in which the process B is performed and to switch from the state in which the process B is performed to the state in which the process A is performed. The PLC **100** controls the device **300B** and the device **300C**, the device **300D** and the device **300E**, and the conveyor CV based on the above instructions.

(17) The structure of the PLC **100** is described in detail below with reference to FIG. 2.

(18) As illustrated in FIG. 2, the PLC **100** includes a communication device **160** that communicates with external devices. The communication device **160** receives instructions from the programmable display **200** and transmits control commands to the devices **300**.

(19) The PLC **100** also includes a storage **140** that stores a sequence program **110** defining a procedure for controlling the devices **300** and the conveyor CV. The PLC **100** also includes a processor **150** that executes the sequence program **110**. The processor **150** executes the sequence program **110** to control the devices **300** and the conveyor CV.

(20) The structure of the sequence program **110** is described below with reference to FIG. 3.

(21) FIG. 3 illustrates the structure of the sequence program **110** written in the sequential function chart. The sequence program **110** includes steps ST0 to ST7 and transitions TR1 to TR11.

(22) The steps ST0, ST1, ST2, ST3, ST4, and ST5 define control actions for the respective devices **300A**, **300B**, **300C**, **300D**, **300E**, and **300F** illustrated in FIG. 1.

(23) The transition TR1 defines a transition condition for transition of an active state from the step ST0 to the step ST1. Similarly, the transition TR2 defines the transition condition for transition of the active state from the step ST1 to the step ST2. The same applies to the other transitions TR3 to TR11.

(24) The steps ST1 and ST2 and the transitions TR2 and TR3 are included in a block BL1 that defines the action to perform the process A illustrated in FIG. 1. The steps ST3 and ST4 and the transitions TR5 and TR6 are included in a block BL2 that defines the action to perform the process B illustrated in FIG. 1.

(25) In the structure described above, the step ST0 being the initial step is an example of a first execution element that defines the action for the device **300A** as a preceding action. The block BL1 is an example of a second execution element that defines the actions for the devices **300B** and **300C** as subsequent actions performed subsequently to the preceding action. The transition TR1 is an example of a transition condition element that defines the transition condition for transition of the active state from the first execution element to the second execution element.

(26) The blocks BL1 and BL2 are included in a selection sequence SS in which the blocks BL1 and BL2 are activated alternatively. In other words, the block BL2 is an example of a branch execution element that is activated alternatively to the second execution element.

(27) The issues to be solved with the structure according to the present embodiment are described below.

(28) The user may operate the display screen **210** in the programmable display **200** illustrated in FIG. 1 to perform the process B among the process A and the process B. In this case, in FIG. 3, the

steps ST3 and ST4 in the block BL2 are sequentially activated when the transition condition defined in the transition TR4 is satisfied after the step ST0. In contrast, the steps ST1 and ST2 in the block BL1 remain in an inactive state.

(29) With the block BL1 remaining in the inactive state as described above, the devices 300B and 300C that perform the process A stop in FIG. 1. This allows maintenance of the devices 300B and 300C while the process B is being performed. In other words, the selection line CVb arranged parallel to the main line CVA allows maintenance of the devices 300B and 300C without stopping the manufacture of products.

(30) However, at a manufacturing site, a user other than the user who has previously operated the display screen 210 to perform the process B may operate the display screen 210 to perform the process A (hereafter referred to as a switching operation) without noticing the above maintenance being performed.

(31) In this case, the programmable display 200 that receives the switching operation provides, to the PLC 100, an external interrupt instruction IR1 to activate the step ST1 illustrated in FIG. 3. The external interrupt instruction IR1 indicates a request for forcibly activating the step ST1 independently of whether the transition condition defined in the transition TR1 is satisfied.

(32) This causes the steps ST1 and ST2 in the block BL1 to be sequentially activated. In other words, in FIG. 1, the devices 300B and 300C that are under maintenance each receive a control command, and may be activated.

(33) The activation during maintenance may cause failures in the devices 300B and 300C. A technique is thus awaited for preventing the activation of the step ST1 due to the external interrupt instruction IR1 and allowing secure maintenance of the devices 300B and 300C.

(34) Additionally, the user may accidentally start the maintenance of the devices 300B and 300C that perform the process A without operating the display screen 210 illustrated in FIG. 1 to perform the process B. In this case, when the transition condition defined in the transition TR1 is satisfied in FIG. 3, the step ST1 is activated.

(35) When the step ST1 is activated, the device 300B illustrated in FIG. 1 under maintenance may be activated and cause failures in the device 300B and the device 300C. A technique is thus awaited for preventing the activation of the step ST1 due to the transition condition defined in the transition TR1 being satisfied and allowing secure maintenance of the devices 300B and 300C.

(36) The devices 300B and 300C may also be activated by an internal interrupt during maintenance. This is described below.

(37) In FIG. 3, the step ST7 includes an activation code describing the activation of the step ST1. In other words, the step ST7 is an example of a calling execution element in an aspect of the present disclosure.

(38) In this case, when the step ST7 is activated, the above activation code is executed to generate an internal interrupt instruction IR2 requesting the activation of the step ST1. Similarly to the external interrupt instruction IR1, the internal interrupt instruction IR2 indicates a request for forcibly activating the step ST1 independently of whether the transition condition defined in the transition TR1 is satisfied.

(39) Thus, the devices 300B and 300C can be activated when the step ST7 is activated during maintenance of the devices 300B and 300C illustrated in FIG. 1. This may also cause failures in the devices 300B and 300C. A technique is thus awaited for preventing the activation of the step ST1 due to the internal interrupt instruction IR2 and allowing secure maintenance of the devices 300B and 300C.

(40) The external interrupt instruction IR1 and the internal interrupt instruction IR2 are collectively referred to as interrupt instructions. The interrupt instructions are each a request for forcibly activating a specific step independently of whether the transition condition defined in the transition TR1 is satisfied.

(41) The PLC 100 according to the present embodiment has a structure responding to the issues

described above. The structure is described below.

(42) As illustrated in FIG. 2, the storage **140** in the PLC **100** also stores an active state control program **120** and flag data **130**.

(43) The flag data **130** indicates whether activation of the block BL1 illustrated in FIG. 3 is to be permitted or prohibited. The activation of the block BL1 herein refers to activation of the steps ST1 and ST2 included in the block BL1.

(44) The active state control program **120** defines a procedure for controlling, based on the indication in the flag data **130**, whether to permit or prevent the activation of the block BL1 due to (I) the transition condition defined in the transition TR1 being satisfied, (II) the external interrupt instruction IR1, or (III) the internal interrupt instruction IR2.

(45) The user at the manufacturing site can update the flag data **130** using the programmable display **200** illustrated in FIG. 1. In other words, the user can rewrite the flag data **130** as appropriate to indicate that the activation of the block BL1 is permitted or to indicate that the activation of the block BL1 is prohibited by operating the display screen **210** in the programmable display **200**.

(46) The functions implemented by the processor **150** illustrated in FIG. 2 executing the active state control program **120** are described in detail below with reference to FIG. 4.

(47) As illustrated in FIG. 4, the PLC **100** functions as a flag data update receiver **151** that receives an update of the flag data **130**.

(48) As described above, the user can operate the programmable display **200** illustrated in FIG. 1 to update, as appropriate, the flag data **130** indicating whether the block BL1 illustrated in FIG. 3 can be activated. The programmable display **200** instructs the PLC **100** to update the flag data **130** in real time based on the user operation. The flag data update receiver **151** receives the update instruction from the programmable display **200** and updates the flag data **130** based on the instruction.

(49) The PLC **100** also functions as a transition condition determiner **152** that determines whether the transition condition for transition to the above block BL1 is satisfied. In FIG. 3, after an action defined in the step ST0 is performed, the transition condition determiner **152** determines whether the transition condition defined in the transition TR1 is satisfied.

(50) The PLC **100** also functions as an external interrupt receiver **153** that receives the external interrupt instruction IR1. As described above, the programmable display **200** that has received the switching operation indicating a switch from performing the process B to performing the process A provides the external interrupt instruction IR1 to the PLC **100**. The external interrupt receiver **153** receives the external interrupt instruction IR1 from the programmable display **200**.

(51) The PLC **100** also functions as an internal interrupt receiver **154** that receives the internal interrupt instruction IR2. As described above, the internal interrupt instruction IR2 indicates the activation of the block BL1 through execution of the activation code included in the step ST7 illustrated in FIG. 3 when the step ST7 is activated.

(52) The PLC **100** also functions as an executor **156** that performs the steps ST0 to ST7 in the sequence program **110** illustrated in FIG. 3. Performing the steps ST0 to ST7 herein refers to outputting, to the devices **300**, a control command corresponding to each of the actions defined in the steps ST0 to ST7.

(53) The PLC **100** also functions as a preventer **155** that prevents the activation of the block BL1 when the flag data **130** indicates that the activation of the block BL1 is prohibited. The preventer **155** includes a transition preventer **155a**, an external interrupt preventer **155b**, and an internal interrupt preventer **155c**.

(54) The transition preventer **155a** first refers to the latest flag data **130** when the transition condition determiner **152** determines that the transition condition in the transition TR1 is satisfied. When the flag data **130** indicates that the activation of the block BL1 is prohibited, the transition preventer **155a** maintains the active state of the step ST0 illustrated in FIG. 3 to prevent the

activation of the block BL1.

(55) When preventing the activation of the block BL1, the transition preventer **155a** outputs a transition invalidation error ER1 indicating that the activation of the block BL1 is prevented to the programmable display **200**. Upon receiving the transition invalidation error ER1, the programmable display **200** causes the display screen **210** to indicate that the activation of the block BL1 is prevented.

(56) When the external interrupt receiver **153** receives the external interrupt instruction IR1, the external interrupt preventer **155b** first refers to the latest flag data **130**. When the flag data **130** indicates that the activation of block BL1 is prohibited, the external interrupt preventer **155b** maintains the inactive state of the block BL1.

(57) When maintaining the inactive state of the block BL1, the external interrupt preventer **155b** outputs an external interrupt invalidation error ER2 indicating that the external interrupt instruction IR1 is invalidated to the programmable display **200** that is the source of the external interrupt instruction IR1. Upon receiving the external interrupt invalidation error ER2, the programmable display **200** causes the display screen **210** to indicate that the operation to switch from the process B to the process A is invalidated.

(58) When the internal interrupt receiver **154** receives the internal interrupt instruction IR2, the internal interrupt preventer **155c** first refers to the latest flag data **130**. When the flag data **130** indicates that the activation of the block BL1 is prohibited, the internal interrupt preventer **155c** maintains the inactive state of the block BL1.

(59) The internal interrupt preventer **155c** outputs an internal interrupt invalidation error ER3 indicating that the internal interrupt instruction IR2 is invalidated to the programmable display **200** and the executor **156**. Upon receiving the internal interrupt invalidation error ER3, the programmable display **200** causes the display screen **210** to indicate that the activation of the block BL1 is prevented and that the step ST7 is not performed. Upon receiving the internal interrupt invalidation error ER3, the executor **156** does not perform the step ST7.

(60) When the flag data **130** indicates that the activation of block BL1 is permitted, the transition preventer **155a**, the external interrupt preventer **155b**, and the internal interrupt preventer **155c** each transmit an instruction to permit the activation of the block BL1 to the executor **156**.

(61) The executor **156** activates the block BL1 when the executor **156** receives the instruction to permit the activation of the block BL1 from (i) the transition preventer **155a**, (ii) the external interrupt preventer **155b**, or (iii) the internal interrupt preventer **155c**.

(62) The external interrupt receiver **153** and the internal interrupt receiver **154** are collectively referred to as interrupt receivers. The external interrupt preventer **155b** and the internal interrupt preventer **155c** are collectively referred to as interrupt preventers. The interrupt receivers receive the interrupt instructions. When an interrupt receiver receives an interrupt instruction, an interrupt preventer first refers to the latest flag data **130**. When the flag data **130** indicates that the activation of block BL1 is prohibited, the interrupt preventer maintains the inactive state of the block BL1 and outputs an interrupt invalidation error to the programmable display **200** and the executor **156**.

(63) The operation of the PLC **100** for determining whether to activate the execution element is described below with reference to FIGS. 5 to 7. The execution element that is a target of the determination as to whether to perform the activation is hereafter referred to as a target execution element. As illustrated in FIG. 3, the block BL1 is the target execution element in the present embodiment.

(64) In this example, the flag data **130** indicating whether to activate the target execution element is updated to the latest data in advance by the user of the programmable display **200**. In other words, the flag data update receiver **151** receives the update instruction from the programmable display **200** and updates the flag data **130** based on the instruction.

(65) A transition prevention process for preventing activation of the target execution element due to the transition condition being satisfied is described first with reference to FIG. 5.

(66) As illustrated in FIG. 5, the transition condition determiner **152** first determines whether the transition condition defined in the transition **TR1** is satisfied (step **S11**). When the transition condition is not satisfied (No in step **S11**), the processing returns to step **S11**.

(67) When the transition condition is satisfied (Yes in step **S11**), the transition preventer **155a** determines whether the latest flag data **130** indicates that the activation of the block **BL1** being the target execution element is permitted (step **S12**).

(68) When the latest flag data **130** indicates that the activation of the target execution element is prohibited (No in step **S12**), the transition preventer **155a** maintains the active state of the step **ST0** that is the execution element preceding the target execution element to prevent the activation of the target execution element. The transition preventer **155a** also outputs the transition invalidation error **ER1** indicating that the activation of the target execution element is prevented to the programmable display **200** (step **S13**).

(69) In contrast, when the latest flag data **130** indicates that the activation of the target execution element is permitted (Yes in step **S12**), the executor **156** activates the block **BL1** that is the target execution element (step **S14**). After steps **S13** and **S14**, the processing returns to step **S11** again.

(70) An external interrupt prevention process for preventing the activation of the target execution element due to the external interrupt instruction **IR1** is now described with reference to FIG. 6.

(71) As illustrated in FIG. 6, the external interrupt receiver **153** constantly determines whether the external interrupt instruction **IR1** is generated (step **S21**). When no external interrupt instruction **IR1** is generated (No in step **S21**), the processing returns to step **S21**.

(72) When the external interrupt instruction **IR1** is generated (Yes in step **S21**), the external interrupt receiver **153** receives the external interrupt instruction **IR1**. The external interrupt preventer **155b** then determines whether the latest flag data **130** indicates that the activation of the block **BL1** being the target execution element is permitted (step **S22**).

(73) When the latest flag data **130** indicates that the activation of the target execution element is prohibited (No in step **S22**), the external interrupt preventer **155b** maintains the inactive state of the block **BL1** that is the target execution element. The external interrupt preventer **155b** also outputs the external interrupt invalidation error **ER2** indicating that the external interrupt instruction **IR1** is invalidated to the programmable display **200** (step **S23**).

(74) In contrast, when the latest flag data **130** indicates that the activation of the target execution element is permitted (Yes in step **S22**), the executor **156** activates the block **BL1** that is the target execution element (step **S24**). After steps **S23** and **S24**, the processing returns to step **S21** again.

(75) An internal interrupt prevention process for preventing the activation of the target execution element due to the internal interrupt instruction **IR2** is now described with reference to FIG. 7.

(76) As illustrated in FIG. 7, the internal interrupt receiver **154** constantly determines whether the internal interrupt instruction **IR2** is generated (step **S31**). When no internal interrupt instruction **IR2** is generated (No in step **S31**), the processing returns to step **S31**.

(77) When the internal interrupt instruction **IR2** is generated (Yes in step **S31**), the internal interrupt receiver **154** receives the internal interrupt instruction **IR2**. The internal interrupt preventer **155c** then determines whether the latest flag data **130** indicates that the activation of the block **BL1** being the target execution element is permitted (step **S32**).

(78) When the latest flag data **130** indicates that the activation of the target execution element is prohibited (No in step **S32**), the internal interrupt preventer **155c** maintains the inactive state of the block **BL1** that is the target execution element. The internal interrupt preventer **155c** also outputs the internal interrupt invalidation error **ER3** indicating that the internal interrupt instruction **IR2** is invalidated to the programmable display **200** and the executor **156** (step **S33**).

(79) In contrast, when the latest flag data **130** indicates that the activation of the target execution element is permitted (Yes in step **S32**), the executor **156** activates the block **BL1** that is the target execution element (step **S34**). After steps **S33** and **S34**, the processing returns to step **S31** again.

(80) As described above, with the flag data **130** indicating that the activation of the target execution

element is prohibited, the structure according to the present embodiment can maintain the inactive state of the target execution element when the external interrupt instruction IR1 is generated, when the internal interrupt instruction IR2 is generated, or when the transition condition to the target execution element is satisfied. This allows, for example, secure maintenance of the devices **300** that are controlled based on the description for the target execution element, or secure debugging of the target execution element. After maintenance or debugging, the flag data **130** may be updated to permit the activation of the target execution elements.

(81) The embodiment are described above. The embodiment may be modified in the following manners.

(82) In the structure according to the above embodiment, the flag data **130** sets whether to activate the block BL1. The second execution element in an aspect of the present disclosure is not limited to the block BL1. The target execution element to be the target of activation setting may be any execution element. In other words, the flag data **130** can set whether to activate any of the execution elements included in the sequence program **110**, and the processes illustrated in FIG. 5 to 7 can be performed on the execution element for which the activate setting is set. The flag data **130** may thus include execution element identification information identifying the execution element for which the activation setting is set and activation information indicating whether to activate the execution element, and update the activation information as appropriate.

(83) FIG. 1 illustrates the programmable display **200** as an example of the human machine device in an aspect of the present disclosure. The human machine device may not be the programmable display **200**, but may be any device that can instruct, when receiving an operation to indicate the update of the flag data **130** from the user, the flag data update receiver **151** to update the flag data **130** based on the operation. The human machine device may be a computer called an engineering tool that provides the user with an environment for creating the sequence program **110**.

(84) FIG. 3 illustrates an example sequence program **110** written in the sequential function chart. The sequence program **110** may be written in any language that includes execution elements and transition condition elements, other than the sequential function chart.

(85) The active state control program **120** illustrated in FIG. 2 may be installed in an existing PLC to allow the PLC to function as the PLC **100** according to the embodiment. The active state control program **120** may be distributed through a communication network, or may be stored in a non-transitory computer-readable recording medium and distributed.

(86) The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full range of equivalents to which such claims are entitled.

REFERENCE SIGNS LIST

(87) **100** Programmable logic controller **110** Sequence program **120** Active state control program **130** Flag data **140** Storage **150** Processor **151** Flag data update receiver **152** Transition condition determiner **153** External interrupt receiver **154** Internal interrupt receiver **155** Preventer **155a** Transition preventer **155b** External interrupt preventer **155c** Internal interrupt preventer **156** Executor **160** Communication device **200** Programmable display (human machine device) **210** Display screen **300**, **300A**, **300B**, **300C**, **300D**, **300E**, **300F** Device **400** Sequence control system CL Communication line CV Conveyor CVa Main line CVb Selection line ST0 Step (first execution element) ST1 Step ST2 Step ST3 Step ST4 Step ST5 Step ST6 Step ST7 Step (calling execution element) TR1 Transition (transition condition element) TR2 Transition TR3 Transition TR4 Transition TR5 Transition TR6 Transition TR7 Transition TR8 Transition TR9 Transition TRi0 Transition TR11 Transition BL1 Block (second execution element) BL2 Block (branch execution

element) SS Selection sequence IR1 External interrupt instruction IR2 Internal interrupt instruction
ER1 Transition invalidation error ER2 External interrupt invalidation error ER3 Internal interrupt
invalidation error

Claims

1. A programmable logic controller for executing a sequence program, the sequence program including a first execution element defining a preceding action, a second execution element defining a subsequent action to be performed subsequently to the preceding action, and a transition condition element defining a transition condition for transition of an active state from the first execution element to the second execution element, the programmable logic controller comprising: a flag data update receiver to receive an update of flag data indicating whether activation of the second execution element is permitted or prohibited; an interrupt receiver to receive an interrupt instruction indicating a request for activating the second execution element independently of whether the transition condition is satisfied; an interrupt preventer to refer to latest flag data when the interrupt receiver receives the interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, an inactive state of the second execution element and output an interrupt invalidation error indicating that the interrupt instruction is invalidated; and an executor to activate the second execution element when the interrupt receiver receives the interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.
2. The programmable logic controller according to claim 1, wherein the interrupt receiver includes an external interrupt receiver to receive an external interrupt instruction being the interrupt instruction from outside the programmable logic controller, the interrupt preventer includes an external interrupt preventer to refer to the latest flag data when the external interrupt receiver receives the external interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, the inactive state of the second execution element and output, as the interrupt invalidation error, an external interrupt invalidation error indicating that the external interrupt instruction is invalidated, and the executor activates the second execution element when the external interrupt receiver receives the external interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.
3. The programmable logic controller according to claim 2, wherein the sequence program includes a branch execution element included, together with the second execution element, in a selection sequence, and the branch execution element is activated alternatively to the second execution element, the external interrupt instruction requests switching the active state from the branch execution element to the second execution element, and the external interrupt preventer refers to the latest flag data when the external interrupt receiver receives the external interrupt instruction, and maintains, when the flag data indicates that the activation of the second execution element is prohibited, the active state of the branch execution element to prevent the activation of the second execution element.
4. The programmable logic controller according to claim 1, wherein the sequence program includes a calling execution element different from the first execution element and the second execution element and including an activation code describing the activation of the second execution element independently of whether the transition condition is satisfied, the interrupt receiver includes an internal interrupt receiver to receive, when the calling execution element is activated, an internal interrupt instruction being the interrupt instruction indicating the activation of the second execution element upon execution of the activation code, the interrupt preventer includes an internal interrupt preventer to refer to the latest flag data when the internal interrupt receiver receives the internal interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, the inactive state of the second execution element and output, as

the interrupt invalidation error, an internal interrupt invalidation error indicating that the internal interrupt instruction is invalidated, and the executor activates the second execution element when the internal interrupt receiver receives the internal interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.

5. The programmable logic controller according to claim 1, further comprising: a transition condition determiner to determine whether the transition condition defined in the transition condition element is satisfied after the preceding action is performed; and a transition preventer to refer to the latest flag data when the transition condition determiner determines that the transition condition is satisfied, and maintain, when the flag data indicates that the activation of the second execution element is prohibited, the active state of the first execution element to prevent the activation of the second execution element, wherein the executor activates the second execution element when the transition condition determiner determines that the transition condition is satisfied and the latest flag data indicates that the activation of the second execution element is permitted.

6. A sequence control system, comprising: the programmable logic controller according to claim 1; and a human machine interface to receive an operation to indicate the update of the flag data, and to instruct the flag data update receiver to update the flag data based on the operation.

7. A non-transitory computer readable medium storing an active state control program for a programmable logic controller to execute a sequence program, the sequence program including a first execution element defining a preceding action, a second execution element defining a subsequent action to be performed subsequently to the preceding action, and a transition condition element defining a transition condition for transition of an active state from the first execution element to the second execution element, the active state control program causing the programmable logic controller to function as: a flag data update receiver to receive an update of flag data indicating whether activation of the second element is permitted or prohibited; an interrupt receiver to receive an interrupt instruction indicating a request for activating the second execution element independently of whether the transition condition is satisfied; an interrupt preventer to refer to latest flag data when the interrupt receiver receives the interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, an inactive state of the second execution element and output an interrupt invalidation error indicating that the interrupt instruction is invalidated; and an executor to activate the second execution element when the interrupt receiver receives the interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.

8. The programmable logic controller according to claim 2, wherein the sequence program includes a calling execution element different from the first execution element and the second execution element and including an activation code describing the activation of the second execution element independently of whether the transition condition is satisfied, the interrupt receiver includes an internal interrupt receiver to receive, when the calling execution element is activated, an internal interrupt instruction being the interrupt instruction indicating the activation of the second execution element upon execution of the activation code, the interrupt preventer includes an internal interrupt preventer to refer to the latest flag data when the internal interrupt receiver receives the internal interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, the inactive state of the second execution element and output, as the interrupt invalidation error, an internal interrupt invalidation error indicating that the internal interrupt instruction is invalidated, and the executor activates the second execution element when the internal interrupt receiver receives the internal interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.

9. The programmable logic controller according to claim 3, wherein the sequence program includes a calling execution element different from the first execution element and the second execution element and including an activation code describing the activation of the second execution element

independently of whether the transition condition is satisfied, the interrupt receiver includes an internal interrupt receiver to receive, when the calling execution element is activated, an internal interrupt instruction being the interrupt instruction indicating the activation of the second execution element upon execution of the activation code, the interrupt preventer includes an internal interrupt preventer to refer to the latest flag data when the internal interrupt receiver receives the internal interrupt instruction, and to maintain, when the flag data indicates that the activation of the second execution element is prohibited, the inactive state of the second execution element and output, as the interrupt invalidation error, an internal interrupt invalidation error indicating that the internal interrupt instruction is invalidated, and the executor activates the second execution element when the internal interrupt receiver receives the internal interrupt instruction and the latest flag data indicates that the activation of the second execution element is permitted.

10. The programmable logic controller according to claim 2, further comprising: a transition condition determiner to determine whether the transition condition defined in the transition condition element is satisfied after the preceding action is performed; and a transition preventer to refer to the latest flag data when the transition condition determiner determines that the transition condition is satisfied, and maintain, when the flag data indicates that the activation of the second execution element is prohibited, the active state of the first execution element to prevent the activation of the second execution element, wherein the executor activates the second execution element when the transition condition determiner determines that the transition condition is satisfied and the latest flag data indicates that the activation of the second execution element is permitted.

11. The programmable logic controller according to claim 3, further comprising: a transition condition determiner to determine whether the transition condition defined in the transition condition element is satisfied after the preceding action is performed; and a transition preventer to refer to the latest flag data when the transition condition determiner determines that the transition condition is satisfied, and maintain, when the flag data indicates that the activation of the second execution element is prohibited, the active state of the first execution element to prevent the activation of the second execution element, wherein the executor activates the second execution element when the transition condition determiner determines that the transition condition is satisfied and the latest flag data indicates that the activation of the second execution element is permitted.

12. The programmable logic controller according to claim 4, further comprising: a transition condition determiner to determine whether the transition condition defined in the transition condition element is satisfied after the preceding action is performed; and a transition preventer to refer to the latest flag data when the transition condition determiner determines that the transition condition is satisfied, and maintain, when the flag data indicates that the activation of the second execution element is prohibited, the active state of the first execution element to prevent the activation of the second execution element, wherein the executor activates the second execution element when the transition condition determiner determines that the transition condition is satisfied and the latest flag data indicates that the activation of the second execution element is permitted.

13. The programmable logic controller according to claim 9, further comprising: a transition condition determiner to determine whether the transition condition defined in the transition condition element is satisfied after the preceding action is performed; and a transition preventer to refer to the latest flag data when the transition condition determiner determines that the transition condition is satisfied, and maintain, when the flag data indicates that the activation of the second execution element is prohibited, the active state of the first execution element to prevent the activation of the second execution element, wherein the executor activates the second execution element when the transition condition determiner determines that the transition condition is satisfied and the latest flag data indicates that the activation of the second execution element is

permitted.

14. A sequence control system, comprising: the programmable logic controller according to claim 2; and a human machine interface to receive an operation to indicate an update of the flag data, and to instruct the flag data update receiver to update the flag data based on the operation.

15. A sequence control system, comprising: the programmable logic controller according to claim 3; and a human machine interface to receive an operation to indicate an update of the flag data, and to instruct the flag data update receiver to update the flag data based on the operation.

16. A sequence control system, comprising: the programmable logic controller according to claim 4; and a human machine interface to receive an operation to indicate an update of the flag data, and to instruct the flag data update receiver to update the flag data based on the operation.

17. A sequence control system, comprising: the programmable logic controller according to claim 5; and a human machine interface to receive an operation to indicate an update of the flag data, and to instruct the flag data update receiver to update the flag data based on the operation.

18. A sequence control system, comprising: the programmable logic controller according to claim 13; and a human machine interface to receive an operation to indicate an update of the flag data, and to instruct the flag data update receiver to update the flag data based on the operation.
